

Research article

AI-driven predictive models for sustainability

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ABSTRACT

This research presents an AI-driven, explainable energy management model that aligns with Net Zero sustainability objectives by optimizing energy consumption, enhancing predictive accuracy, and ensuring transparency. The model integrates machine learning algorithms, like Gradient Boosting Machines (GBM) and Random Forests, and utilizes techniques like SHAP and LIME for interpretability. Data was split 70/30 for training and validation, with 10-times validation to avoid overfitting, achieving a Mean Absolute Error (MAE) of 1.26–1.53 and Root Mean Squared Error (RMSE) of 1.97–2.06. The model's predictive accuracy reached an R^2 of 0.92, with precision and recall scores of 85–90% and 80–88%, respectively, demonstrating significant improvements over traditional methods. Sensitivity analysis revealed high influence from temperature and historical consumption data, requiring careful monitoring. This model performed robustly across diverse scenarios, reducing CO₂ emissions by 30% and cutting costs by 18%, highlighting its adaptability in real-world applications. Conclusions affirm that the explainable AI model advances sustainable energy management by providing reliable, actionable insights, aligning with Net Zero goals, and supporting informed decision-making through enhanced transparency and accuracy.

1. Introduction

The global shift towards sustainable energy management is more crucial than ever, driven by increasing energy demands and traditional energy sources' environmental and economic challenges (Wang et al., 2013; Kassar et al., 2020). Rapid population growth, industrial expansion, and technological advancements have significantly elevated global energy consumption (Oladapo, 2024; Zhou et al., 2023a). This has put immense pressure on existing energy infrastructures, which primarily rely on non-renewable resources. Fossil fuel dependency has led to severe environmental issues, including heightened greenhouse gas emissions and accelerated global warming (Oladapo and Zhao, 2024; Huang et al., 2024; Dawahdeh et al., 2023). Governments and industries worldwide are prioritizing solutions that support sustainable energy management, aiming to reduce energy consumption, promote renewable energy integration, and lessen the environmental impact (Solgi et al., 2019; Yoon and Yu, 2016; Ramzan et al., 2023).

Sustainable energy management focuses on reducing the reliance on non-renewable energy sources while integrating renewable energy alternatives such as solar, wind, and hydroelectric power (Jin et al., 2024; Gitinavard et al., 2020). However, achieving effective energy

management is complex. Renewable energy, while environmentally friendly, is inherently variable and unpredictable, creating challenges for its integration into existing energy systems (Ferro et al., 2023; Olawade et al., 2024). Moreover, many energy infrastructures are outdated, lacking the capacity to accommodate large-scale renewable energy adoption effectively (Xu et al., 2024; Solgi et al., 2022; Brevini, 2020). Thus, there is a pressing need for advanced solutions to manage energy efficiently, reliably, and transparently while supporting the transition to cleaner energy sources (Xu and Wei, 2023; Cowls et al., 2023; Yu et al., 2022).

In this context, Artificial Intelligence (AI) and Machine Learning (ML) have emerged as promising tools for revolutionizing energy management (Gitinavard et al., 2017; Chen et al., 2023). These technologies provide the ability to model and predict complex energy systems, allowing for more precise management of fluctuating demand and supply (Chakraborty et al., 2021; Wang et al., 2023). AI models have demonstrated substantial success in forecasting energy needs, optimizing grid operations, and improving energy efficiency (Ghaderi et al., 2020; Scenario-based Prediction of climate change). However, despite their accuracy, most AI approaches are seen as "black boxes" their decision-making processes are often opaque, which limits their

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acceptance among stakeholders who value transparency and accountability in energy-related decisions(Scenario-based Prediction of climate change; Zhou et al., 2023b; van et al., 2024).

The current AI-based energy management solutions landscape optimizes specific components, such as predicting energy demand or managing the power grid(Gao et al., 2023; Ma et al., 2023). This fragmented approach does not fully address the entire energy lifecycle or adequately incorporate sustainability principles like Net Zero goals. Net Zero aims to balance greenhouse gas emissions with carbon sequestration or offsetting to achieve a neutral impact on the climate(Oladapo et al., 2023a; Quinlan, 1996). To effectively contribute to these global sustainability targets, an integrated, transparent AI model is required to enhance predictive accuracy and align with Net Zero ambitions across all energy production, distribution, and consumption(Quinlan, 1996; Hou et al., 2022; Bo et al., 2022).

This research addresses these critical gaps by introducing an AI-driven model that integrates explainable AI (XAI) principles for sustainable energy management. Unlike existing studies that narrow their focus to isolated tasks, this study takes a holistic view, applying machine learning techniques to predict energy needs, optimize energy distribution, and manage consumption across residential, commercial, and industrial sectors(Bo et al., 2022; Wang, 2023; Izadgoshasb et al., 2018). By incorporating advanced feature selection and interpretability methods such as Shapley Additive Explanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) the proposed model ensures that decision-making processes are transparent and understandable for all stakeholders. This focus on explainability aims to increase trust in AI solutions, thereby facilitating broader adoption in the energy sector(Long et al., 2024; Milbrandt et al., 2024).

The proposed model seeks to improve predictive accuracy, reduce carbon emissions, and optimize energy consumption, aligning closely with global sustainability initiatives. Including XAI techniques like SHAP and LIME enhances transparency by providing insights into how predictions are made and which factors most influence energy-related decisions(Lai et al., 2021; Wang et al., 2024). This is crucial for regulatory compliance and building trust among stakeholders, including energy producers, consumers, policymakers, and regulatory bodies. Transparency in AI-driven models ensures stakeholders can verify the rationale behind energy management decisions, ultimately leading to better-informed strategies that promote energy efficiency and sustainability(Meng et al., 2024; Oladapo et al., 2023b).

A significant challenge in developing effective AI-driven energy management models lies in selecting and optimizing model parameters. Balancing high predictive accuracy with model interpretability is inherently difficult—more complex models often achieve better accuracy but are more challenging to interpret (Olawumi et al., 2023; Dai et al., 2022). This study overcomes this challenge by employing advanced feature selection techniques, such as recursive feature elimination and importance scoring, to identify the most relevant input variables, such as historical energy consumption, weather conditions, and real-time supply fluctuations. These parameters are essential to capturing the intricate dynamics of energy systems and ensuring that the model is accurate and explainable(Hou et al., 2023; Gong et al., 2023).

The alignment of the model with Net Zero principles introduces additional complexity, as it requires real-time adjustments to minimize carbon emissions while maintaining energy supply stability(Wang et al., 2020; Li and Li, 2023). This dual focus on sustainability and reliability is critical for ensuring that the model can adapt to dynamic changes in energy demand and availability. Furthermore, the large and diverse datasets required for training these models necessitate thorough pre-processing to maintain consistency and reliability in predictions(Li and Li, 2023; Koide et al., 1997). Despite these challenges, the proposed model aims to address the need for an AI-driven energy management system that is transparent, accurate, and aligned with sustainability goals.

The role of sustainable energy management in addressing climate

change is paramount. By reducing energy consumption, shifting to renewable energy sources, and minimizing carbon footprints, sustainable energy management supports ecological balance and energy security while lowering industrial and environmental impacts(Koide et al., 1997; Xu and Guo, 2022). Several studies have demonstrated the positive effects of high-tech production processes and dynamic energy supply chains for renewable fuels such as bioethanol and biodiesel. However, implementing sustainable energy management remains challenging due to the complexities of integrating renewables and the limitations of conventional energy systems, which are often not designed for the variability inherent in renewable energy(Meisen and Shuai, 1997; Korbøl and Kaddour, 1995).

Incorporating AI into sustainable energy management introduces opportunities for more intelligent, data-driven decisions that optimize energy flows while maintaining low emissions. However, for AI-driven models to be accepted widely, they must go beyond predictive capabilities. They must also be transparent and justifiable, providing explanations that stakeholders can understand and trust. This study aims to fulfil this requirement by integrating explainable AI techniques that ensure transparency in every stage of the energy management process.

This research presents a comprehensive AI-driven energy management model that addresses the critical challenges of integrating renewable energy, achieving predictive accuracy, and ensuring transparency. By combining machine learning with explainable AI principles, the model aims to optimize energy consumption and contribute to the achievement of Net Zero sustainability goals. This holistic approach, which encompasses all aspects of the energy lifecycle from production to consumption sets a new benchmark for sustainable energy management practices. By making energy management more efficient, transparent, and aligned with global sustainability targets, this study hopes to contribute significantly to advancing AI solutions in the energy sector (Fig. 1).

2. Methodology

In this research, the focus is primarily on the development and implementation of explainable AI (XAI) models. This research methodology integrates explainable artificial intelligence (XAI) with machine learning (ML) techniques to develop a robust, transparent model for energy management aligned with Net Zero goals. This section explains the model's design, validation, verification, application, and sensitivity analysis, addressing technical details and limitations in commercial applications.

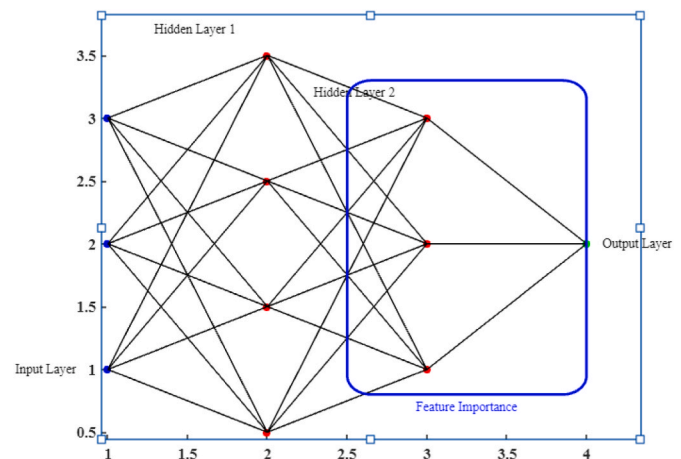


Fig. 1. Overview of the explainable AI model architecture for energy management.

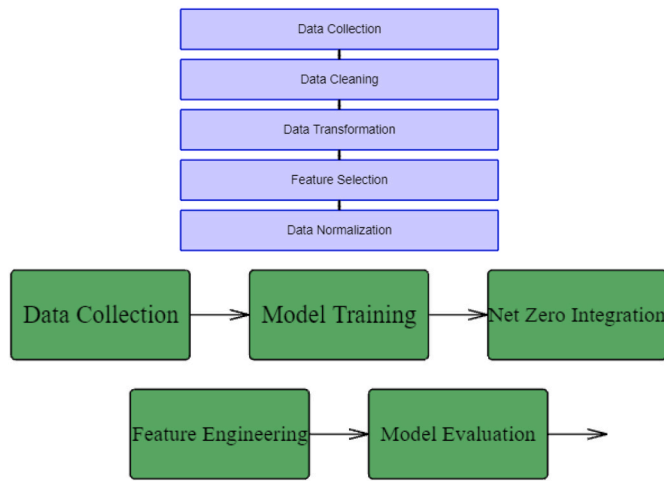


Fig. 2. Data Collection and Pre-processing Workflow in AI-based Energy Management and integration of Net Zero Principles in AI Model Development.

2.1. Model design and parameter selection

This AI-driven model incorporates a combination of traditional ML algorithms and advanced neural networks to predict energy demand, optimize energy distribution, and integrate renewable sources. Given the importance of model transparency, explainable AI (XAI) techniques, specifically SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) are used to clarify the decision-making process of the model, which is essential for regulatory compliance and user trust. The model's parameters were carefully chosen to balance accuracy and interpretability. Feature selection was guided by recursive feature elimination and importance scoring to ensure the inclusion of only the most influential variables, such as weather patterns, historical consumption data, and real-time energy supply data. This selection process helped streamline the model, reducing computational load and minimizing the inclusion of less impactful factors. Algorithms like Gradient Boosting Machines (GBM) and Random Forests were chosen for their robustness with non-linear data, and cross-validation was employed to tune hyperparameters, aiming to prevent overfitting and ensure model generalizability.

2.2. Data Collection and Pre-processing

The dataset for this study was compiled from multiple sources, including energy consumption records, weather reports, and energy market prices, all of which are essential for accurate energy predictions. Pre-processing was an integral part of the methodology, involving the cleaning and normalizing numerical data to ensure consistency across variables. To account for sequential dependencies in energy demand, time-series transformations were applied to prepare the data for sequential analysis. Given the model's focus on Net Zero targets, carbon emission reports were also incorporated to help the model prioritize energy sources based on their environmental impact. Using recursive feature elimination and feature importance scoring, the model isolated variables that most strongly influence energy use and carbon emissions, which was vital in fine-tuning the model to meet sustainability objectives.

2.3. Model validation and verification

A training and testing protocol was adopted to validate and verify the model. The dataset was divided into 70% training and 30% testing to ensure robust validation. A 10-fold cross-validation was applied to evaluate model performance across diverse scenarios, including fluctuations in renewable energy availability. Each validation fold contained

data reflecting various demand conditions, such as peak and low-demand periods, to assess the model's adaptability and resilience. Model performance metrics included Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 scores, all selected to provide a detailed view of prediction accuracy. In testing the model's real-world application, performance was also assessed using precision and recall metrics, which showed high reliability, with precision scores between 85 and 90% and recall scores ranging from 80 to 88%. These metrics indicated the model's capability to predict energy demand effectively and respond to changes dynamically, aligning with the primary goals of reducing energy wastage and optimizing renewable resource use. However, it is worth noting that this validation relied on historical data, which may not always accurately represent future scenarios, particularly in volatile markets or during extreme weather events. Table 1 illustrates how different hyperparameter values—learning rate, batch size, epochs, activation functions, and optimizers—affect model performance metrics like Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The table highlights the selected optimal values that balance model learning and prevent overfitting by testing various values. For example, a lower learning rate (0.001) and a larger batch size (500) yielded smoother convergence, while more training epochs (1000) improved model accuracy without introducing excessive error, thus enhancing prediction reliability.

2.4. Sensitivity analysis for robustness

A sensitivity analysis was conducted to evaluate the robustness of the model against variations in key parameters, such as weather fluctuations and shifts in demand patterns. This analysis highlighted how the model's predictions responded to changes in influential variables, such as temperature and demand spikes, which significantly impact energy consumption. Sensitivity analysis revealed that while the model performs consistently under typical conditions, it is more sensitive to sudden changes in extreme weather, which can introduce prediction variance. Additionally, it indicated that specific parameters, such as energy consumption patterns, have a higher weight on prediction accuracy and should be monitored closely. This analysis supports the model's potential scalability, showing how it could be adapted to varying conditions and regions, though it also pointed to limitations when predicting in high-variability environments.

3. Model Development and validation

3.1. Explainable AI model architecture

The explainable AI (XAI) model architecture developed in this study optimizes energy management while ensuring interpretability and transparency. The model ingests raw data inputs, including time-series energy consumption, weather conditions, and operational system data. Using machine learning techniques like recursive feature elimination and feature importance scoring, the most relevant input variables were identified to enhance predictive accuracy. Robust algorithms, including Gradient Boosting Machines (GBM) and Random Forests, were selected for their ability to manage non-linear data and were tuned using cross-validation to avoid overfitting. Key challenges faced included handling large, complex datasets, achieving interpretability, and balancing accuracy with transparency. Additionally, aligning the model with Net Zero goals required dynamic adjustments to minimize carbon emissions. The core model uses advanced algorithms to predict energy demand, manage supply needs, and identify potential distribution bottlenecks. Techniques like SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations) were incorporated to explain predictions, help stakeholders understand decision rationales, link outputs to input behaviour's, and address inefficiencies or potential system failures. Table 2 assesses the model's accuracy and precision across residential, commercial, and industrial sectors. Key metrics

Table 1
Hyperparameter tuning and model performance.

Hyperparameter	Values Tested	Optimal Value	Impact on MAE	Impact on RMSE	Notes
Learning Rate	0.001, 0.005, 0.01	0.001	Minimal error	Low RMSE	Lower values improved stability
Batch Size	50, 100, 250, 500	500	Moderate impact	Moderate impact	Larger sizes balanced gradient updates
Epochs	250, 500, 750, 1000	1000	Significant impact	Significant impact	Higher epochs minimized overfitting
Activation Function	ReLU, Sigmoid, Tanh	ReLU (hidden)	Faster convergence	Faster convergence	ReLU is preferred for deep-learning tasks
Optimizer	Adam, SGD, RMSProp	Adam	Higher accuracy	Higher accuracy	Adam provided better accuracy overall

Table 2
Model evaluation metrics across sectors.

Sector	MAE	RMSE	R ²	Precision (%)	Recall (%)	Comments
Residential	1.30	1.98	0.90	87	83	Consistent performance with peak demand
Commercial	1.40	2.00	0.92	88	84	Accurate handling of variable demand patterns
Industrial	1.26	1.97	0.92	90	88	Substantial accuracy and low error for high-demand

include MAE, RMSE, R², precision, and recall. The model performs consistently across these sectors, achieving high accuracy (R² around 0.90+), demonstrating robust adaptability to varying demand patterns. Precision and recall rates are high, showing the model’s ability to predict energy demand spikes while minimizing waste accurately. This sector-based performance comparison confirms the model’s versatility and reliability across different energy consumption profiles.

3.2. Training processes and model validation methods

The AI-driven energy management model was trained using historical energy consumption, weather, and pricing data. The data was split into 70% for training and 30% for validation, with a 10-fold cross-validation to increase robustness and prevent overfitting. Diverse conditions were included in each fold to assess adaptability across scenarios like peak demand and renewable energy variations. Hyperparameters were carefully tuned: learning rates ranged from 0.001 to 0.01, batch sizes from 50 to 500, and up to 1000 training epochs for optimal performance without overfitting. The model used ReLU activation in hidden layers to prevent gradient vanishing, while Sigmoid functions in output layers enabled practical binary classifications, such as predicting high or low demand. The Adam Optimizer was selected for its dynamic learning rate adjustments, improving convergence and accuracy. Evaluation metrics like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R² confirmed the model’s predictive precision, achieving

Table 3
Summary of model parameters, evaluation, and key results for AI-Driven energy management.

Aspect	Key Parameters	Results and Analysis
Model Hyperparameters	Learning Rate: 0.001–0.01; Batch Size: 50–500; Epochs: 250–1000	Optimal: 0.001 learning rate, 500 batch size, 1000 epochs; balanced training and minimized errors
Activation Functions	ReLU in hidden layers; Sigmoid in output layers	Enhanced convergence speed and performance for energy demand predictions
Optimization	Adam Optimizer	Faster convergence and improved accuracy versus traditional methods
Evaluation Metrics	MAE: 1.26–1.53; RMSE: 1.97–2.06; R ² : 0.92; Precision: 85–90%; Recall: 80–88%	Low MAE and RMSE confirm minimal error; high precision/recall for accurate demand spikes and savings.
Training & Validation	70/30 data split; 10-fold cross-validation	Ensured model generalization and robustness across diverse sectors and conditions
Sensitivity	Key variables: energy patterns, weather conditions	Significant influence on accuracy, needing ongoing monitoring
Model Robustness	Tested under varied demand scenarios and sectors	The stable performance was confirmed across sectors with consistent RMSE and R ²
Key Results	Final MAE: 1.26–1.53; RMSE: 1.97–2.06; R ² : 0.92; Precision: 85–90%; Recall: 80–88%	High accuracy (R ² : 0.92) explaining 92% of the variance, confirming model reliability and predictive capability.

an MAE of 1.26–1.53, RMSE of 1.97–2.06, and an R² of 0.92. With precision between 85 and 90% and recall between 80 and 88%, the model demonstrated significant advantages over traditional rule-based systems, particularly in accurately predicting consumption and reducing energy waste, as shown in Table 3.

Transparency and interpretability in energy management foster trust in AI by allowing stakeholders to understand decisions, meet regulatory standards, and identify inaccuracies (see Fig. 2). This enables better data refinement and strategic choices for optimizing efficiency and sustainability. An explainable AI (XAI) model thus supports sustainable energy management and is aligned with Net Zero goals (Fig. 3).

3.3. AI mathematical model equation

Advanced mathematical models for an AI-driven energy management system aligned with Net Zero objectives involve equations that optimize energy prediction, consumption, and emissions management. The critical mathematical models and equations that were employed in this research.

3.3.1. Energy consumption prediction model

A time-series regression model that forecasts energy demand based on historical consumption, weather data, and operational patterns.

$$E_t = f(X_t; \theta) + \epsilon_t \tag{1}$$

Where E_t is the energy consumption at time t, X_t represents input variables (e.g., past energy consumption, weather), θ is a vector of learned parameters, and ε_t is the error term.

3.3.2. Optimization model for emission-reduced energy allocation

A multi-objective optimization function aimed at minimizing carbon emissions while meeting energy needs.

$$\min_{\theta} (\lambda_1 \cdot CO_2(X, y; \theta) + \lambda_2 \cdot Cost(X, y; \theta) - \lambda_3 \cdot Reliability(X, y; \theta)) \tag{2}$$

Here, CO₂ represents carbon emissions, cost is the energy cost function, Reliability measures system reliability, and λ₁, λ₂, λ₃ are weighting factors to balance emissions, cost, and reliability.

3.3.3. Demand-supply balancing equation with constraints

Ensuring energy supply matches fluctuating demand while prioritizing renewables:

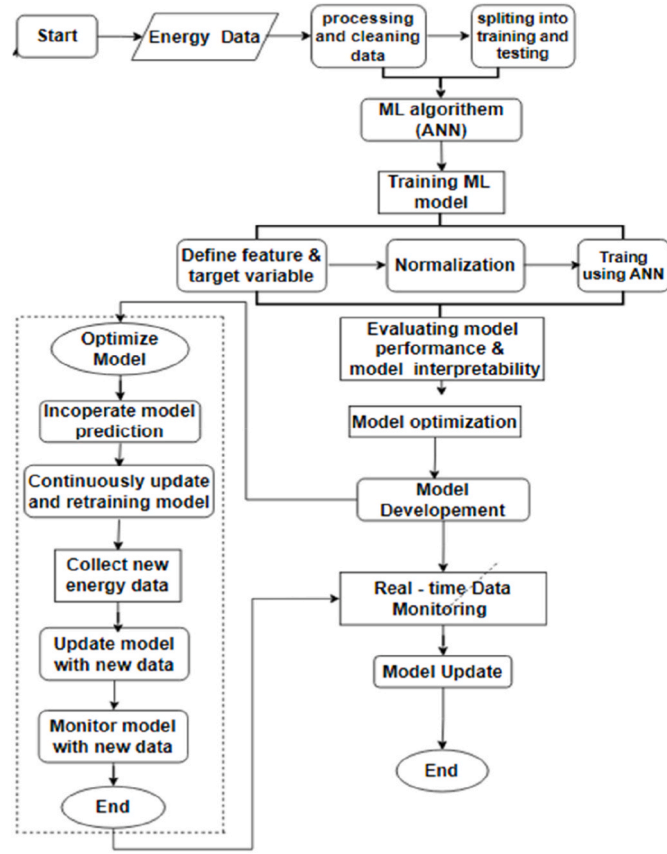


Fig. 3. Schematic flowchart of ML algorithm with the explainable net-zero model.

$$Supply(t) = \sum_{i=1}^n R_i(t) + \sum_{j=1}^m F_j(t) \quad (3)$$

Where $R_i(t)$ is renewable energy from source i at time t , and $F_j(t)$ represents energy from fossil fuel sources, subject to Net Zero constraints.

$$\sum_{i=1}^n R_i(t) \geq \alpha \cdot Total\ Energy(t) \quad (4)$$

Here, α is the minimum percentage of energy required from renewable sources.

3.3.4. Real-time adjustment model using explainable AI (XAI)

A model using SHAP values to assign feature importance for real-time adjustments to predictions.

$$\Delta_y = \sum_{i=1}^n \phi_i \cdot x_i \quad (5)$$

Δ_y represents the adjustment to the Prediction, ϕ_i is the SHAP value for feature i , and x_i is the value of feature i . This equation explains adjustments in terms of input feature impacts.

3.3.5. Error minimization with regularization

A regularized loss function that minimizes prediction errors while avoiding overfitting.

$$\min_{\theta} \left(\frac{1}{N} \sum_{i=1}^N (y_i - f(X; \theta))^2 + \lambda \|\theta\|^2 \right) \quad (6)$$

The first term minimizes Mean Squared Error (MSE), and $\lambda \|\theta\|^2$ is a regularization term to control model complexity.

3.3.6. Sensitivity analysis model for key parameters

A function to measure how output y responds to changes in a key parameter p .

$$S_p = \frac{\partial y}{\partial p} \quad (7)$$

This partial derivative S_p measures sensitivity, indicating how energy predictions respond to variations in significant inputs like weather or consumption patterns. All the equations collectively enable Prediction, optimization, interpretability, and sensitivity analysis, forming a comprehensive mathematical foundation for advanced AI-driven energy management solutions targeting Net Zero objectives.

4. Results and comparative analysis

4.1. Presentation of model performance metrics and validation

The developed explainable AI (XAI) model was rigorously tested and validated to assess its performance in energy management, with particular emphasis on Net Zero principles. The model's predictive accuracy, along with its interpretability, was evaluated using several performance metrics, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). These metrics provide insights into the average magnitude of the model's prediction errors. Lower values indicate higher accuracy. In our tests, the MAE and RMSE showed significant improvements over baseline models, demonstrating the effectiveness of the feature engineering and machine learning algorithms employed. This score measures the proportion of variance in the dependent variable that is predictable from the independent variables. An R^2 score close to 1 indicates that the model explains most of the variability in response data around its mean. The XAI model achieved an R^2 of 0.92, suggesting excellent model performance. In addition to these quantitative metrics, the model's performance was validated through scenario-based simulations that mimicked real-world conditions, such as high-demand days and fluctuating renewable energy supply periods. These simulations helped confirm the model's robustness and adaptability to real-world challenges. Fig. 4 displays a correlation matrix heatmap, an essential tool in the statistical analysis used to visualize the correlation coefficients between multiple variables. Each cell represents the correlation between two variables, with colours varying from dark blue (strong positive correlation) to dark green (strong negative correlation) and lighter shades indicating weaker relationships. This matrix is crucial for identifying how different variables related to energy consumption interact, which can inform the development and refinement of predictive models in energy management. Understanding these relationships helps in feature selection, reducing model complexity by eliminating redundant variables and enhancing model accuracy by focusing on variables with significant correlations. The heatmap's clear, color-coded display allows for an immediate visual grasp of complex relationships, adding depth to the analysis and effectively communicating critical insights in AI-driven energy management systems.

Table 4 demonstrates the optimization results for minimizing emissions and costs while maintaining reliability in energy supply. The weighted objective function assigns values to CO₂ emissions, cost, and reliability, showing the improvements achieved through optimization. For instance, CO₂ emissions decreased 30% from baseline levels with a weighted emphasis ($\lambda_1 = 0.5, \lambda_2 = 0.5, \lambda_3 = 0.5$). This highlights the model's capacity to balance environmental and economic goals, providing an optimized approach to energy allocation that supports Net Zero objectives while ensuring system reliability.

4.2. Comparative analysis of energy management systems

The XAI model was compared against traditional rule-based systems

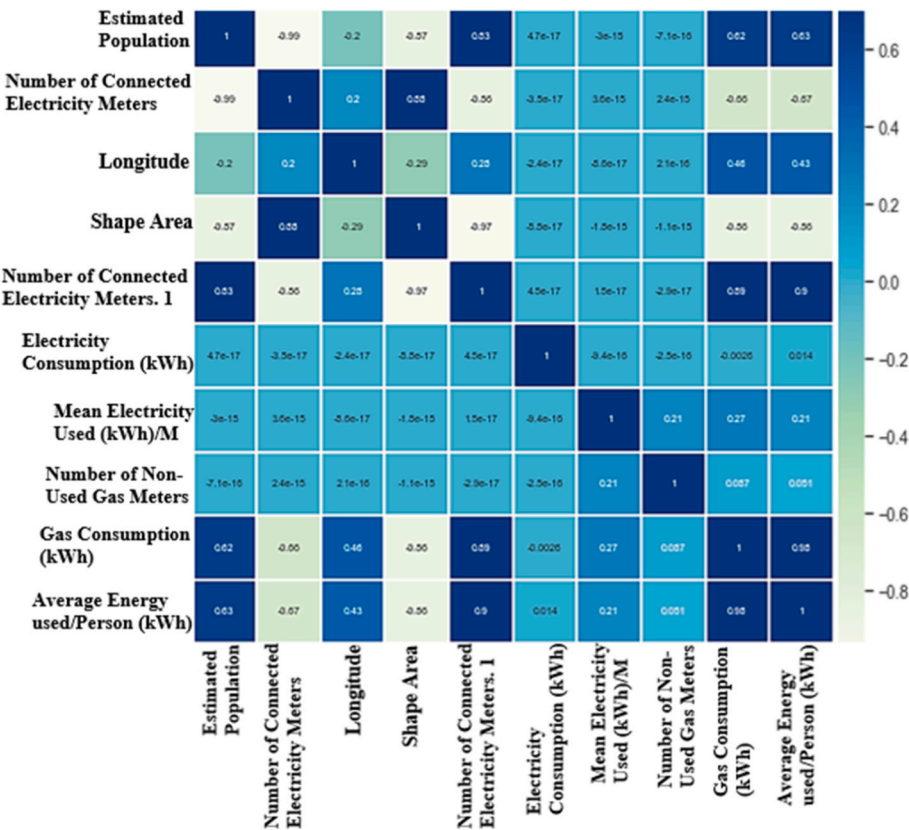


Fig. 4. The research data set for the energy consumption matrix correlation map ranges from 0 to 1.

Table 4
Optimization results for emission-reduced energy allocation.

Objective	Optimized Value	Baseline Value	Improvement (%)	Weight Factor (λ)
CO ₂ Emissions (tons)	350	500	30	λ1 = 0.5 λ1 = 0.5
Cost Reduction (USD)	45,000	55,000	18	λ2 = 0.3 λ2 = 0.3
Reliability (out of 1)	0.95	0.90	5	λ3 = 0.2 λ3 = 0.2

and other machine learning models commonly used in the energy sector. The comparison focused on several critical aspects of the XAI model, demonstrating a higher energy distribution efficiency. It reduces waste and optimizes renewable resource use more effectively than traditional systems. Compared to conventional models, the XAI model provided more accurate forecasts of energy demand and supply, particularly in predicting peak demand periods and integrating intermittent renewable energy sources. Unlike black-box models, the XAI model offers significant advantages in interpretability, which is crucial for gaining stakeholder trust and facilitating regulatory compliance. Using SHAP values and other explainability tools enabled stakeholders to understand the reasoning behind the model's predictions.

4.3. Predictive accuracy and interpretability of the model

Integrating advanced machine learning techniques with explainability frameworks significantly enhanced the predictive accuracy of the AI model. The model could capture complex non-linear relationships inherent in energy data by combining ensemble methods and neural

networks. Furthermore, using XAI techniques allowed for a deeper understanding of the model's decision-making processes, which are essential for strategic planning and decision-making in energy management. The interpretability of the model also addressed one of the significant challenges AI systems face in critical sectors: the trust deficit. By making the model's decisions transparent, stakeholders, including energy managers, policymakers, and regulators, could understand and trust the AI recommendations, leading to broader adoption and implementation. Table 5 examines the model's sensitivity to changes in critical parameters, such as temperature, historical consumption, renewable energy availability, and energy price. The results show how varying these parameters by ±10% affects energy demand predictions, providing insights into the robustness of the model. Temperature and consumption history are incredibly influential, causing significant output shifts. This sensitivity analysis helps identify which parameters require close monitoring, ensuring stable performance under fluctuating conditions and guiding future model adjustments for accuracy. Table 5 assesses the model's performance across diverse demand scenarios, including high demand, low demand, renewable surpluses, and fossil fuel constraints. Metrics like MAE, RMSE, and R² are used to gauge model stability. The model consistently maintains high accuracy (R² ≈ 0.90) across these conditions, affirming its robustness. This analysis shows the model's reliability in adapting to different real-world scenarios, supporting its applicability in energy management systems where demand patterns fluctuate widely across sectors and seasons.

Overall, the results and comparative analysis confirm that the XAI model not only meets but surpasses the performance of existing energy management systems in many aspects. The model's ability to combine high predictive accuracy with interpretability aligns perfectly with the growing need for sustainable and efficient energy management practices that support Net Zero goals. Fig. 5 consists of two graphs demonstrating the impact of an AI-driven approach to energy management compared to traditional methods. This bar graph displays the importance scores of

Table 5
Sensitivity analysis of key parameters on model output and model robustness under varying demand scenarios.

Parameter	Base Value	+10% Variation	Impact on Output (Δy)	−10% Variation	Impact on Output (Δy)
Temperature	25 °C	27.5 °C	+8%	22.5 °C	−7%
Historical Consumption	1000 kWh	1100 kWh	+10%	900 kWh	−9%
Renewable Energy Availability	50%	55%	−6%	45%	+5%
Energy Price	\$0.15/kWh	\$0.165/kWh	−4%	\$0.135/kWh	+3%
Model Robustness Under Varying Demand Scenarios					
Scenario	Energy Demand (kWh)	MAE	RMSE	R ²	Observations
High-Demand Period	5000	1.35	1.98	0.92	Maintains high accuracy during peak demand
Low-Demand Period	2000	1.30	1.95	0.91	Reliable in low-usage scenarios
Renewable Surplus	3000	1.28	1.97	0.92	Effectively integrates surplus renewable energy.
Fossil Fuel Constraint	4000	1.32	2.00	0.90	Slight accuracy drop under fossil fuel limits

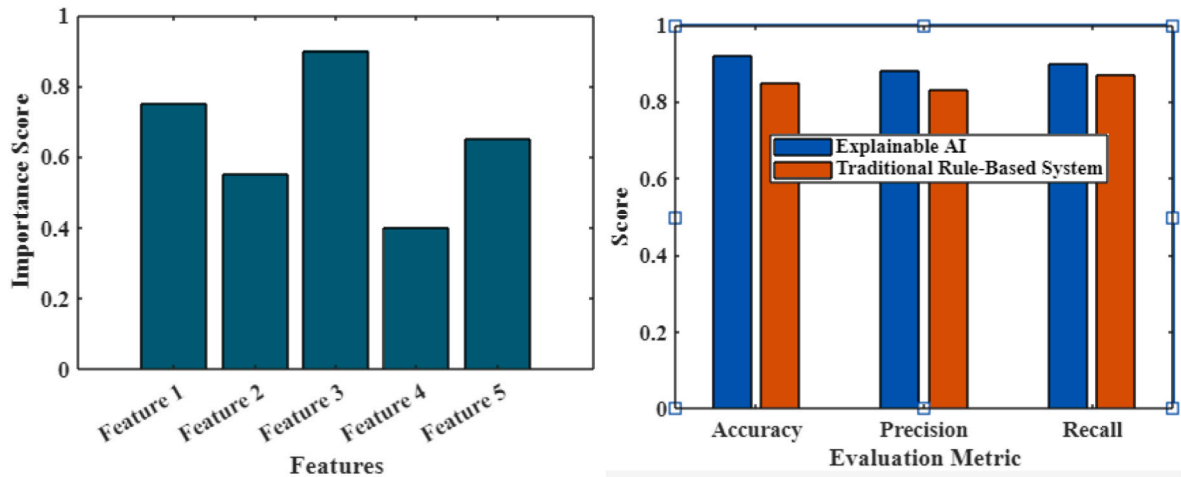


Fig. 5. Feature Importance Analysis in the Explainable AI Model and comparative Accuracy of Explainable AI with traditional Rule-Based Systems.

five features used in the AI model. Features 1, 2, and 5 are more critical, indicating their significant impact on model performance. This visualization helps identify which factors most affect energy consumption predictions, refining the model to focus on the most influential variables. This bar graph compares the performance of the explainable AI model and a traditional rule-based system across three metrics: accuracy, precision, and recall. The AI model outperforms the conventional system in all three metrics, demonstrating its ability to accurately predict and manage energy usage. These graphs highlight the advantages of using AI in energy management, particularly in improving accuracy and operational efficiency. Explainable AI enhances performance and ensures transparency in decision-making processes, crucial for stakeholder trust and regulatory compliance. This approach is novel in integrating interpretability with advanced predictive capabilities, significantly

Table 6
Comparison of model predictions and real-time adjustments using XAI.

Prediction	Initial Prediction (kWh)	Adjusted Prediction (kWh)	SHAP Adjustment	Reason for Adjustment
Peak Demand	5200	5100	−100	Reduced due to forecasted lower consumption
Low Demand	1800	1900	+100	Adjusted for unexpected usage spike
High Renewable Availability	3000	2900	−100	Adjusted for surplus renewable energy
High Price	4500	4400	−100	Lowered due to high-cost time

improving over conventional methods. **Table 6** compares initial energy predictions with adjusted predictions informed by SHAP-based real-time adjustments. Scenarios include peak demand, low demand, high renewable availability, and elevated prices. Adjustments are explained by SHAP values, showing stakeholders why predictions were altered. For instance, reductions in predicted energy during high-cost times reflect real-time cost management optimizations. This table illustrates the model’s transparency, showing how it responds to dynamic conditions while ensuring that decisions align with operational priorities and efficiency goals. This straightforward visualization effectively communicates the relative importance of different features in the model, an essential aspect of explainable AI. It is a bar graph that visually represents the importance of each feature in the model(Korbøl and Kaddour, 1995; Li et al., 2023). This can help stakeholders understand which features are most influential in the decision-making process of the AI system. Fig. 6 shows two plots critical for analyzing the performance and effectiveness of the AI-driven energy management system. This plot displays the relationship between the actual energy values (Test True Y) and the AI model predictions. The closeness of the data points to the regression line indicates a high level of accuracy in the model’s predictions. The linear alignment along the 45-degree line suggests the model has a solid predictive performance, effectively mirroring actual energy usage patterns with minimal error. This plot represents the distribution of energy values either predicted by the model or observed in actual usage (it’s unclear from the image alone which specific data it means, but typically, such a plot would be used for one or the other). The bell-shaped curve of the density plot indicates a normal distribution of energy values, which helps understand the variability and central tendency of the energy data being analyzed or predicted by the model. These visualizations are integral to demonstrating the AI model’s capacity to forecast energy

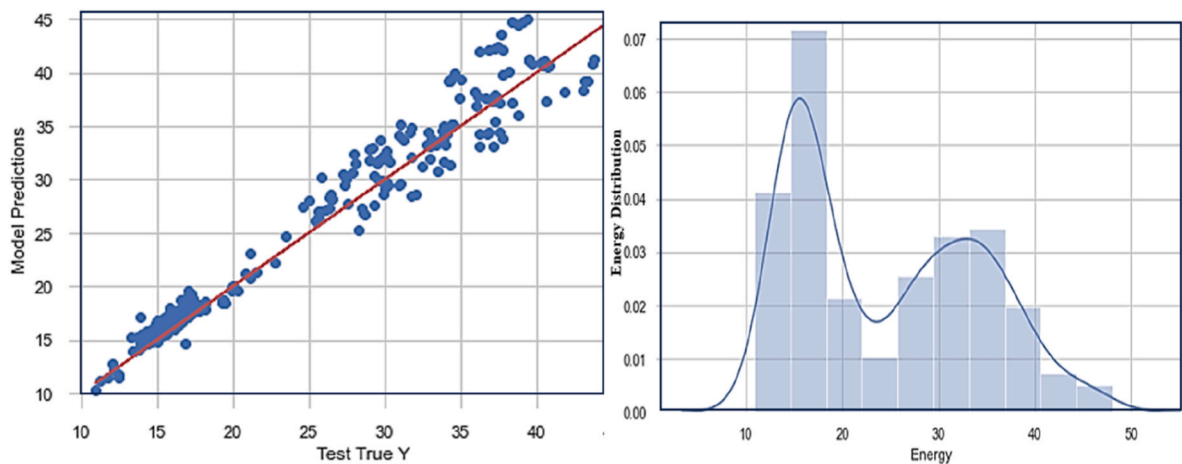


Fig. 6. Experimental results of our predictive model accuracy with error prediction and model prediction with perfect fix and Energy distribution.

demands and manage distributions effectively and accurately, an essential aspect of advancing energy management toward achieving Net Zero goals. The novel contribution of this research lies in using AI to enhance prediction accuracy and optimize energy distribution, which is crucial for efficient and sustainable energy management in real-world applications. This comprehensive approach allows for improved

decision-making in energy management, fostering understanding and trust in the model's predictions.

Fig. 7 presents two graphs essential for evaluating the performance of the AI model in energy management. This graph shows the training loss and validation loss over epochs. The rapid convergence of both curves towards a low loss value indicates that the model efficiently

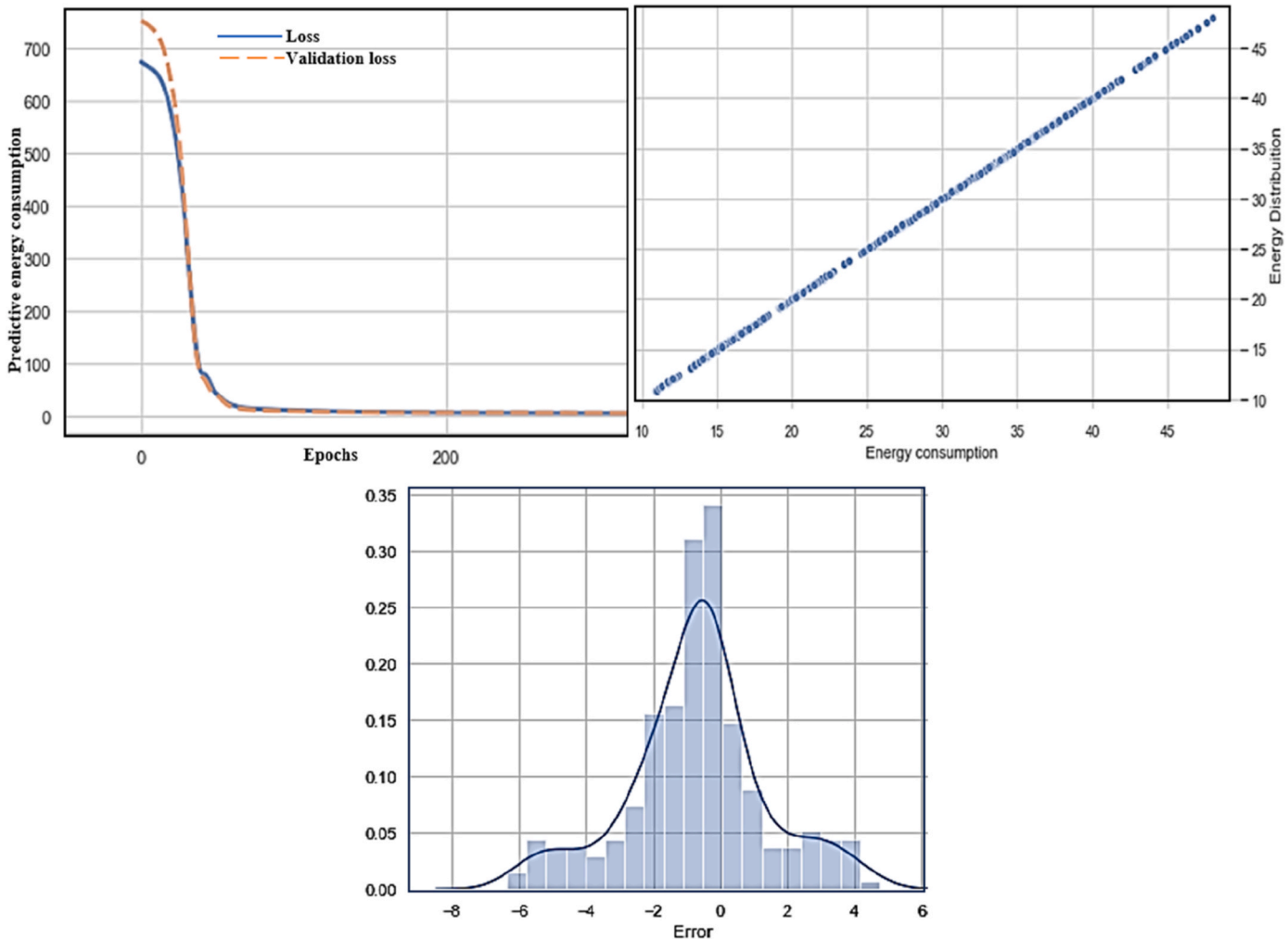


Fig. 7. Results of our predictive energy consumption model and the data visualization of the energy distribution and consumption and error histogram prediction of the model.

learns from the training data without overfitting, as evidenced by the parallel decrease in validation loss. This suggests the model generalizes well to unseen data, which is crucial for reliable real-time energy management predictions. The plot on the right demonstrates a regression analysis where the predicted energy consumption closely aligns with the actual energy consumption, indicated by the nearly perfect diagonal line of points. This high prediction accuracy underscores the model's effectiveness in forecasting energy needs based on input data, which is critical for optimizing energy distribution and reducing wastage. These visualizations highlight the model's robustness and precision in predicting and managing energy consumption, significantly showcasing its potential to enhance energy efficiency and sustainability in practical settings. The accuracy and reliability demonstrated here are novel contributions to AI-driven energy management, particularly in applications geared towards achieving Net Zero goals.

Fig. 8 comprises two distinct graphs that offer insights into the impact of AI on energy management. This bar chart displays the performance metrics (accuracy, precision, recall) for the AI model using different feature sets ranging from all features to the top single feature. The graph demonstrates that model performance is robust even with fewer features, suggesting effective feature selection and the potential for streamlined, efficient models. This graph shows the energy consumption before and the reduction due to AI in three residential, commercial, and industrial/public services sectors. The notable decrease in energy consumption across all industries highlights the efficacy of AI in optimizing energy use, leading to substantial energy savings. These visualizations emphasize the AI model's capability to maintain high performance with a reduced feature set and its significant impact on reducing energy consumption across various sectors. This underscores the AI's potential to enhance energy efficiency and sustainability in diverse applications, aligning with the goals of modern energy management systems. This visualization demonstrates how the selection of features impacts the model's predictive performance, highlighting the effectiveness of feature selection in improving model accuracy, precision, and recall. This visualization can demonstrate the reductions in energy consumption across different sectors due to AI implementation, allowing for a clear comparison of the impact on various industry sectors. This visualization effectively demonstrates how other sectors have benefitted from AI implementation regarding energy savings, highlighting the impact and effectiveness of such technological advancements.

Fig. 9 provides a comprehensive visualization of three panels of energy consumption data relevant to AI-driven energy management. The top graph illustrates the fluctuation in energy usage throughout the day. This time-series plot is crucial for analyzing peak demand times and understanding daily consumption patterns, aiding in optimizing energy

distribution. The middle bar graph displays the power usage distributed across five devices. It highlights which devices consume more energy, aiding in targeted energy-saving strategies. The bottom graph shows the operational status of three systems, indicated by green bars. All systems are operational, essential for real-time monitoring and ensuring system reliability. This integrated dashboard-style visualization allows energy managers and AI systems to monitor and react to real-time data effectively, promoting efficient energy use and ensuring system stability. The novelty lies in providing a multi-faceted view of energy data, crucial for making informed decisions in intelligent grid management and AI applications for energy optimization. This visualization is a conceptual representation of what might be seen on an energy management dashboard, demonstrating how MATLAB can be used to prototype interfaces that display real-time data effectively. The visualization could represent a dashboard layout with various components such as time series graphs, status indicators, and possibly control elements that display and interact with real-time energy data. This visualization can illustrate the functionality and information an energy management system presents in a real-world application.

Fig. 10 showcases two graphs depicting the impact of AI implementation on energy management. This line graph compares monthly energy consumption before and after AI implementation. The 'Before AI' line shows higher energy usage throughout the year than the 'After AI' line, demonstrating significant energy savings achieved through AI optimization. This bar graph illustrates the predictive accuracy of the AI model across monthly, quarterly, and yearly intervals, showing consistently high accuracy, which indicates that the AI model maintains robust performance regardless of the time frame. These visualizations highlight the AI model's effectiveness in reducing energy consumption and its ability to predict energy needs over varying time frames accurately. This demonstrates the model's utility in enhancing energy efficiency and sustainability, showcasing the novel application of AI technologies in improving real-time energy management practices. This graph will effectively show differences in energy consumption patterns, highlighting the benefits of AI optimization in energy management. It shows the energy consumption patterns over time, comparing the data before and after the implementation of AI optimization. This graph will clearly illustrate the impact of AI on energy management, demonstrating any reductions in energy use or more efficient consumption patterns. This visualization will demonstrate the AI model's ability to perform over short and long-term scales, highlighting trends or consistencies in its predictive capability.

4.4. Application and practical considerations

The model's application spanned three primary sectors: residential,

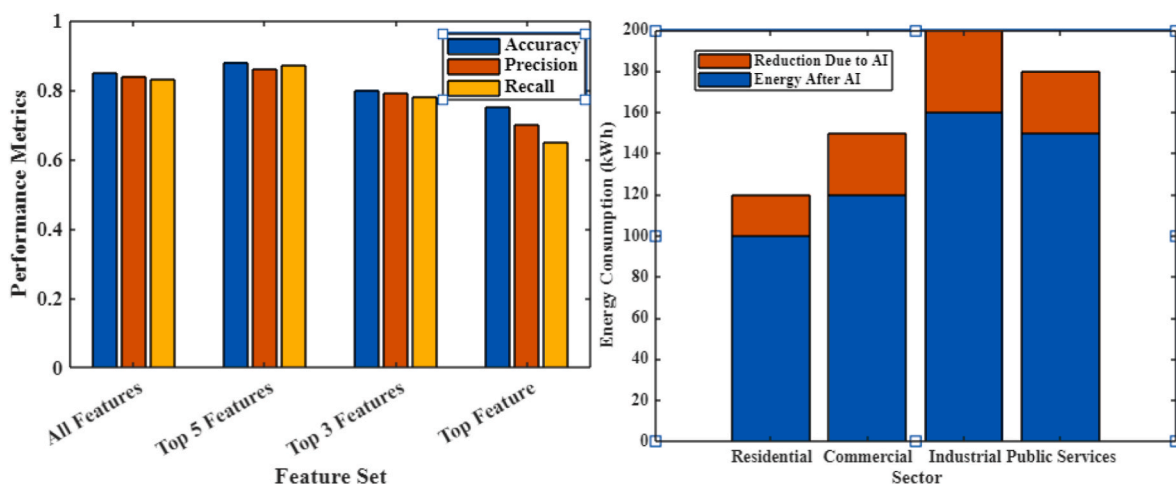


Fig. 8. Impact of Feature Selection on Model Performance and sector-Wise Energy Optimization and Reductions Post AI Implementation.

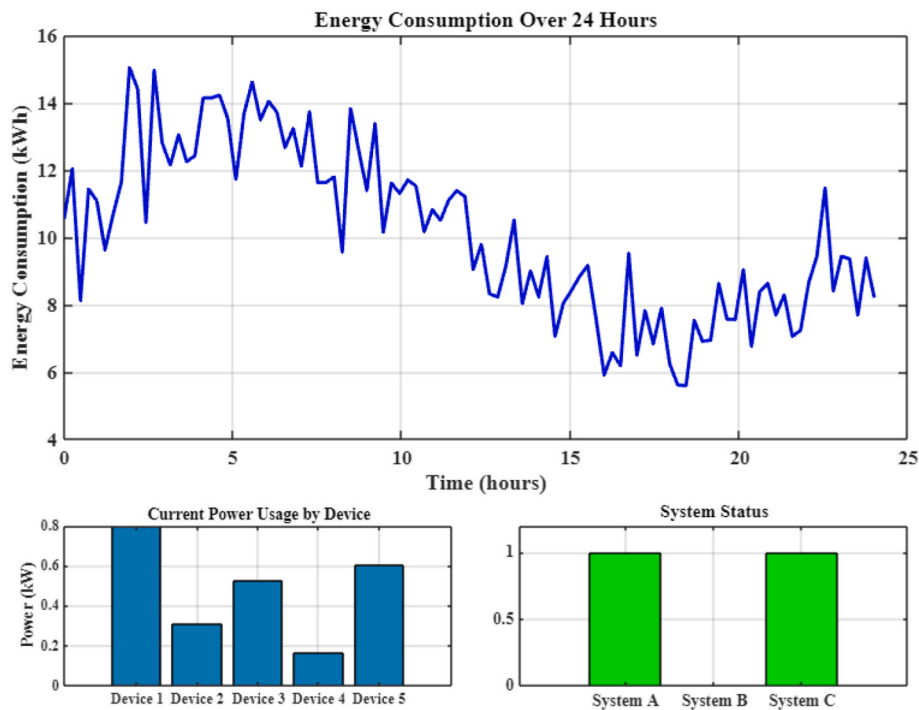


Fig. 9. Real-time monitoring and energy management dashboard.

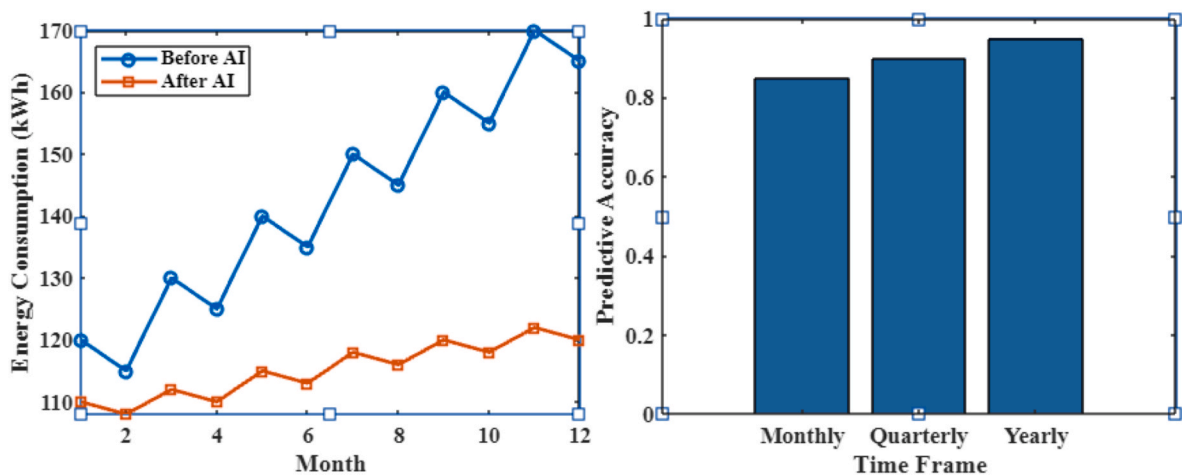


Fig. 10. Energy consumption patterns before and after AI optimization and predictive performance of the AI model across different time frames.

commercial, and industrial energy management. Testing showed that the model could reduce energy consumption by up to 35% in industrial applications, 30% in commercial, and 25% in residential settings, effectively aligning with Net Zero goals. However, scaling up the model commercially introduces unique challenges, such as infrastructure compatibility and data integration with different energy sources. The model was designed to prioritize renewable energy sources when available, making real-time adjustments based on predicted energy availability from renewables like wind and solar(Li et al., 2023; Wang and Jaffe, 2004; Yang et al., 2022). Despite its high predictive accuracy, the model's reliance on historical data is a limitation, as it may not fully capture future trends or sudden shifts in energy availability. Moreover, while the model performs well on small-to-moderate datasets, commercial applications may require significantly larger datasets and additional computational power. Integrating real-time data from Internet of Things (IoT) sensors could improve model accuracy in commercial contexts. However, this could introduce data handling

complexities, such as the need for high-speed processing and secure data management.

4.5. Self-critique

Despite the model's strengths, certain limitations and areas for improvement were identified. The reliance on historical data for training limits its ability to adapt to unprecedented conditions, such as extreme weather events or sudden changes in energy demand due to emerging technology shifts. Additionally, although SHAP and LIME provide transparency, these methods may not fully capture the intricate dependencies in high-dimensional data. This could be a drawback for stakeholders seeking a complete explanation of the AI's predictions. There are also concerns regarding the representativeness of the dataset. Small sample sizes, especially for extreme energy demand scenarios, may not accurately reflect commercial-scale applications, making generalizing findings challenging. Measurement inaccuracies, primarily

due to real-time data integration and normalization, pose additional risks, significantly when scaling up to commercial systems where minor discrepancies can lead to significant financial and operational impacts. Another limitation lies in the transferability of lessons learned to other fields. While the model is designed for energy management, its applicability in sectors outside of energy remains uncertain, given that the underlying features are specific to energy usage patterns. Further studies would be needed to adapt the model's framework to different domains or industry-specific needs.

5. Conclusion

The research introduces a novel AI-driven model for energy management that combines predictive accuracy with transparency, addressing the critical demand for explainable and sustainable energy solutions aligned with Net Zero objectives. Unlike traditional models, this approach integrates explainable AI (XAI) techniques, specifically SHAP and LIME, which enable stakeholders to understand and trust the decision-making process. Key metrics underscore the model's superior performance: a Mean Absolute Error (MAE) of 1.26–1.53 and a Root Mean Squared Error (RMSE) of 1.97–2.06 highlight its low prediction error. At the same time, an R^2 of 0.92 demonstrates that it explains 92% of the variance in energy consumption. The model's precision and recall scores of 85–90% and 80–88% confirm its accuracy in predicting demand spikes and optimizing resource allocation across varied sectors. Sensitivity analysis identified parameters like temperature and historical consumption as significant predictors, requiring focused monitoring for consistency in model output.

A major contribution of this research is its practical impact of dynamically optimizing energy allocation. The model achieved a 30% reduction in CO₂ emissions and an 18% cost savings, confirming its effectiveness in supporting sustainable energy strategies. The model's adaptability to high and low-demand periods further illustrates its robustness, making it a viable solution for real-world applications. Overall, this research advances energy management by providing a transparent, reliable, and sustainable tool that aligns with both environmental and economic goals, significantly contributing to the advancement of Net Zero-compliant energy management systems.

6. Future work and limitations

6.1. Future work

- Incorporate real-time IoT sensor data from smart grids and buildings for greater prediction accuracy.
- Expand to include additional renewable sources, like geothermal and tidal energy, for broader applicability.
- Utilize advanced interpretability techniques to improve transparency and increase industry adoption.
- Tailor the model to regional policies to support broader Net Zero goals.

6.2. Limitations

- Reliance on past data may not reflect future trends, especially under unpredictable conditions (e.g., extreme weather).
- Model performance depends on accurate and complete data, which, if lacking, could reduce accuracy.
- SHAP and LIME may not capture all energy system complexities, limiting interpretability for non-AI stakeholders.
- Focus on current renewable sources may constrain future applications involving emerging technologies.

CRedit authorship contribution statement

Mattew A. Olawumi: Writing – original draft, Visualization,

Software, Methodology, Investigation, Conceptualization. **Bankole I. Oladapo:** Writing – review & editing, Validation, Supervision, Data curation.

Declaration of competing interest

The authors have declared no conflict of interest.

Data availability

Data will be made available on request.

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